

Gen 3

V 5.3

Revised 11/25

Industrial pH Probe

Double junction silver / silver chloride

No Temp

Reads **pH**

Range **0 – 14**

Resolution **+/- 0.001**

Accuracy **+/- 0.002**

Response time **95% in 1s**

Temperature range °C **-5 – 99 °C**

Max pressure **100 PSI**

Max depth **70m (230')**

Connector **SMA or Tinned leads**

Cable length **3 meters (9.8')**

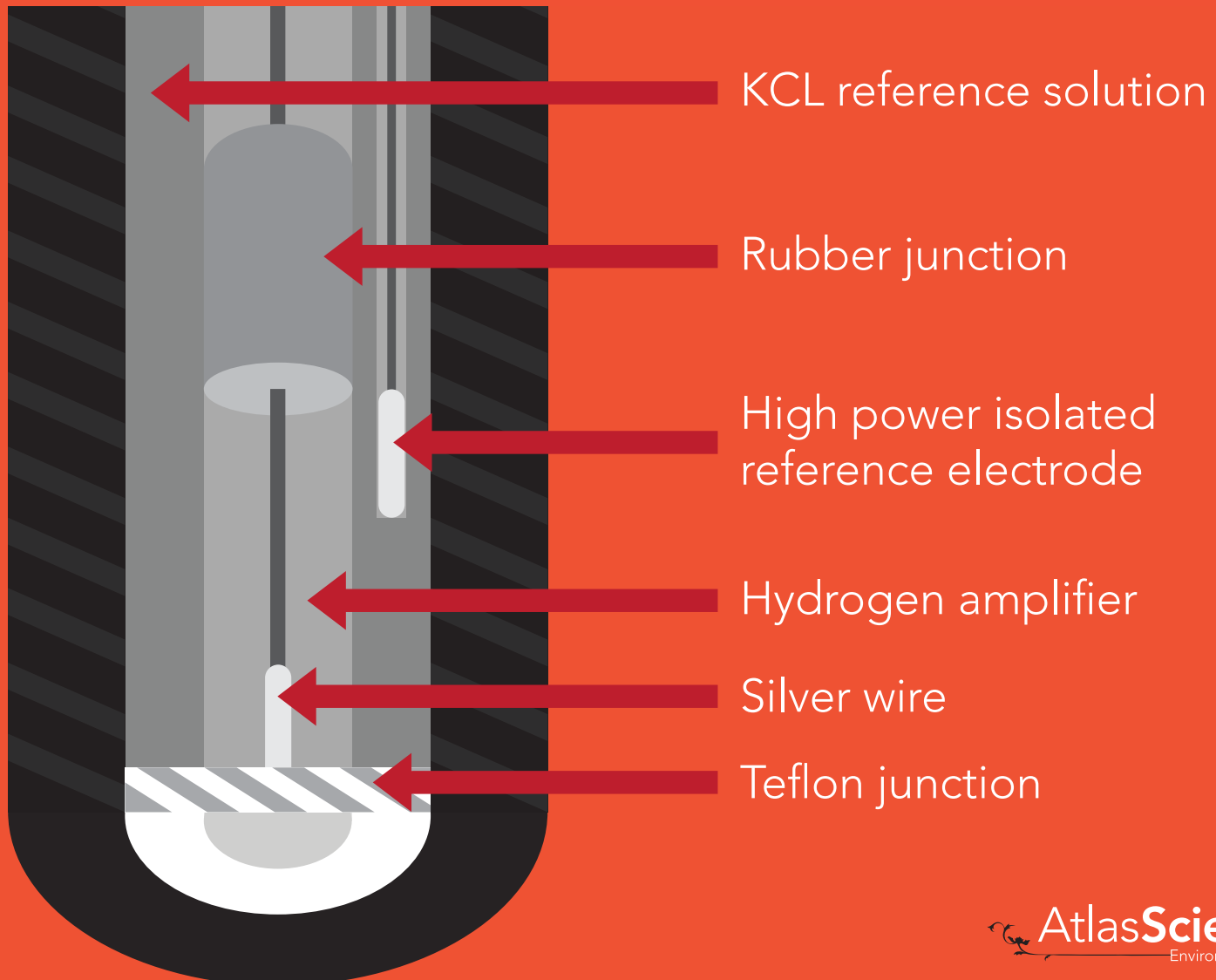
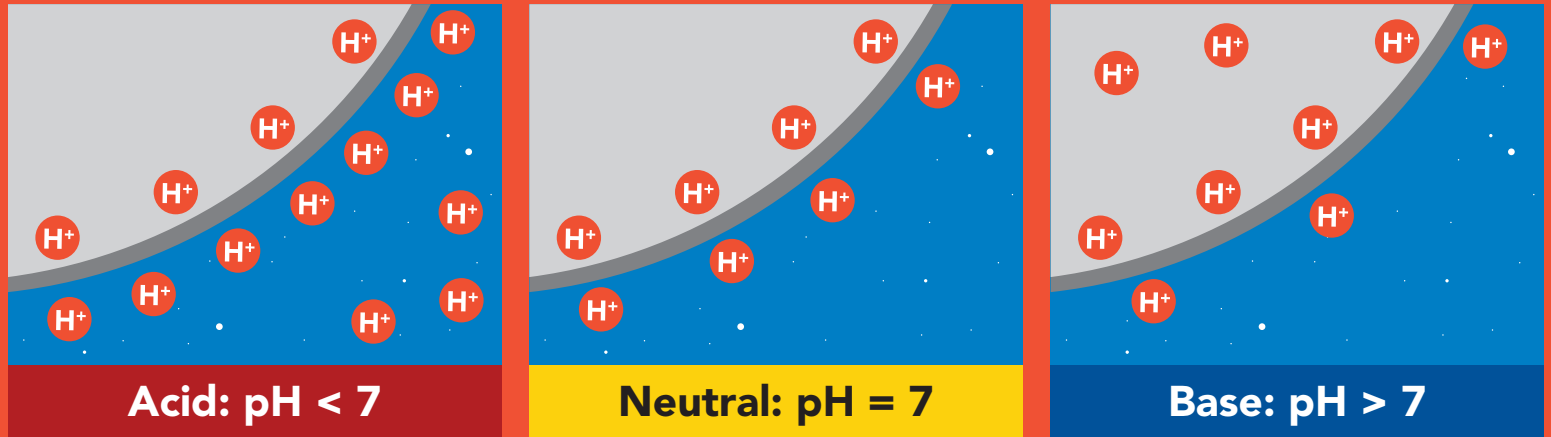
Time before recalibration **~1 Year**

Life expectancy **~4 Years +**



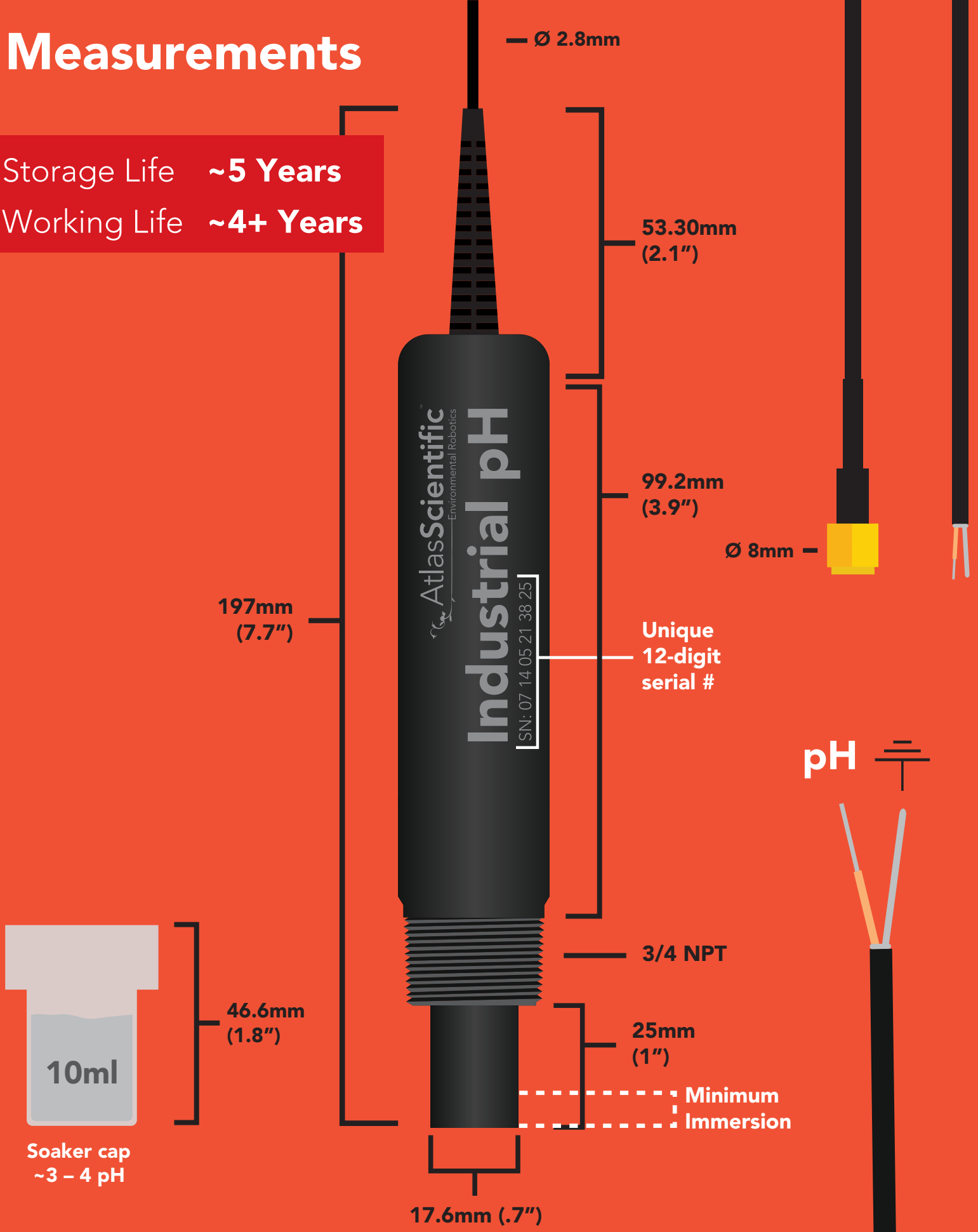
Operating principle

A pH (**potential of Hydrogen**) probe measures the hydrogen ion activity in a liquid. At the tip of a pH probe is a glass membrane. This glass membrane permits hydrogen ions from the liquid being measured to diffuse into the outer layer of the glass, while larger ions remain in the solution. The difference in the concentration of hydrogen ions (outside the probe vs. inside the probe) creates a VERY small current. This current is proportional to the concentration of hydrogen ions in the liquid being measured.



Measurements

Storage Life ~5 Years
Working Life ~4+ Years



NSF/ANSI 51 Compliant

Food Safe

Atlas Scientific LLC, hereby certifies that,

Gen 3 Industrial pH Probe – No Temp
Part # ENV-50-pH-NT

meets the NSF/ANSI Std. 51,
Whether or not they bear the NSF Mark.

✓ **PVC**
NSF-51 Compliant

✓ **Glass**
NSF-51 Compliant

✓ **Teflon**
NSF-51 Compliant

✓ **Delrin®**
NSF-51 Compliant

✓ **EPDM**
NSF-51 Compliant

✓ **Polyethylene**
NSF-51 Compliant

✓ **Titanium Nitride**
NSF-51 Compliant



Specifications

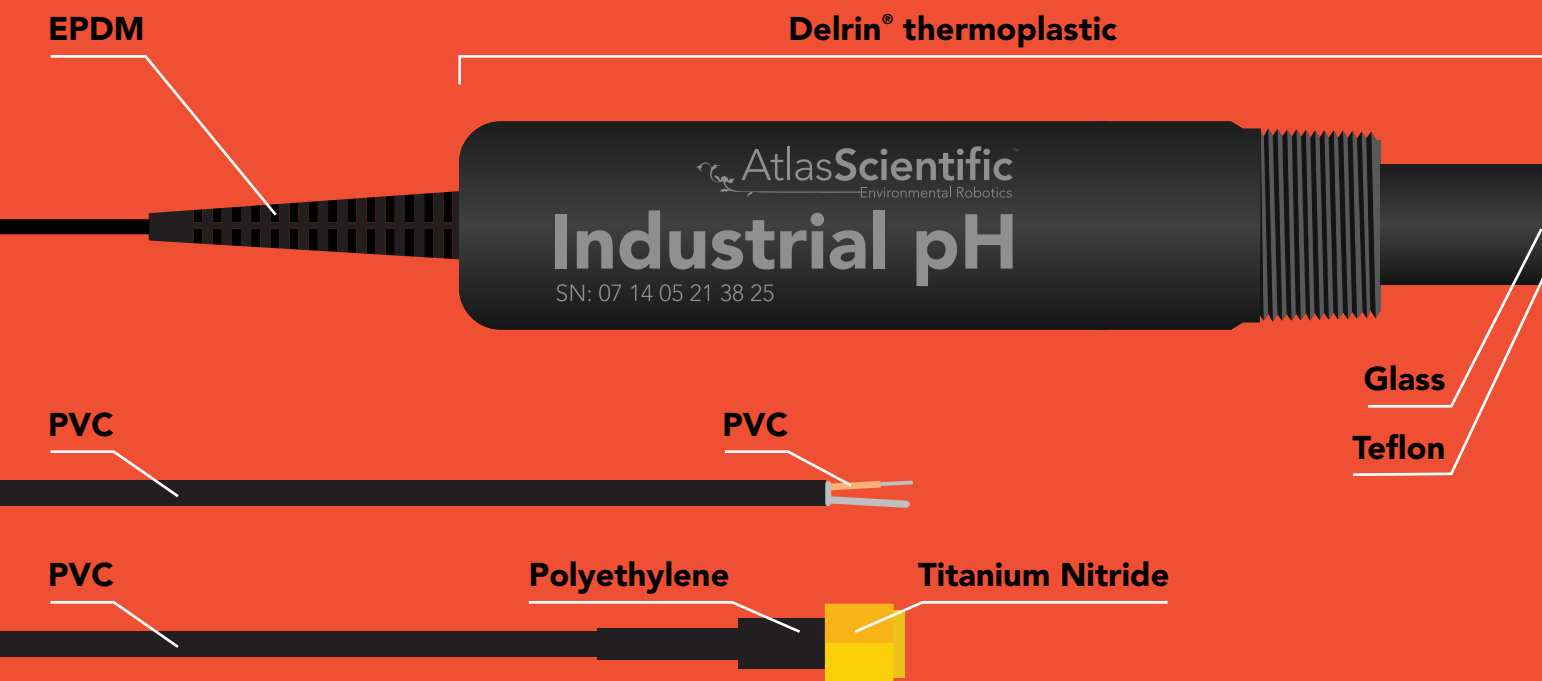
Reference electrode
Body material
Max depth
Cable length
Internal temp. probe
Connector
Weight
Threading
Sterilization
Food safe

Silver / Silver Chloride
Delrin® thermoplastic
70m (230 feet)
3m (9.8 feet)
No
SMA or Tinned Leads
155 grams
(3/4") NPT
Chemical only
Yes

DO NOT BOIL

DO NOT FREEZE

Materials

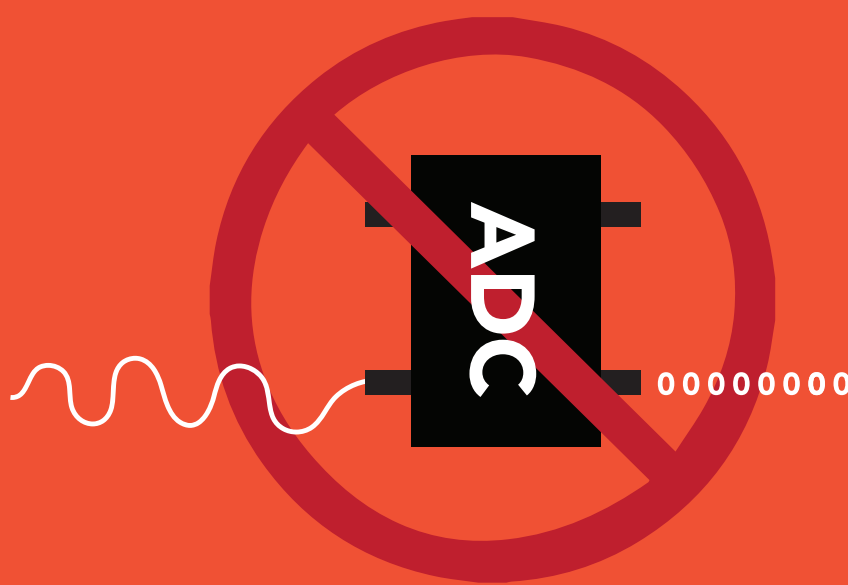


This pH probe can be **fully submerged** in fresh or salt water, up to the SMA connector or tinned leads **indefinitely**.

A pH electrode is a passive device that detects a current generated from hydrogen ion activity. This current (*which can be positive or negative*) is very weak and cannot be detected with a multimeter, or an analog to digital converter. This weak electrical signal can easily be disrupted and care should be taken to only use proper connectors and cables.



Result will **always** read zero.



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The current that is generated from the hydrogen ion activity is the reciprocal of that activity and can be predicted using this equation:

$$E = E^0 + \frac{RT}{F} \ln(\alpha_{H+}) = E^0 - \frac{2.303RT}{F} pH$$

Where **R** is the ideal gas constant.

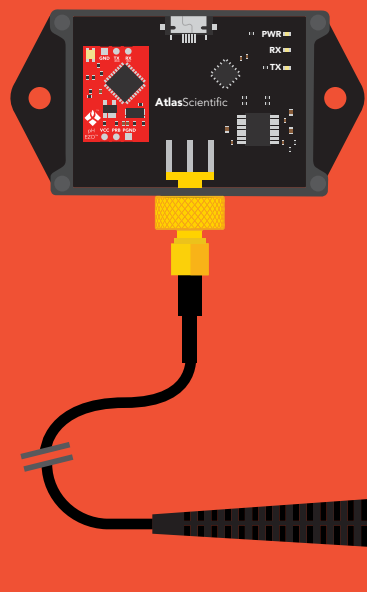
T is the temperature in Kelvin.

F is the Faraday constant.

Because a pH probe is a passive device it can pick up voltages that are transmitted through the solution being measured. This will result in incorrect readings and will slowly damage the pH probe over time. In this instance, proper isolation is required.

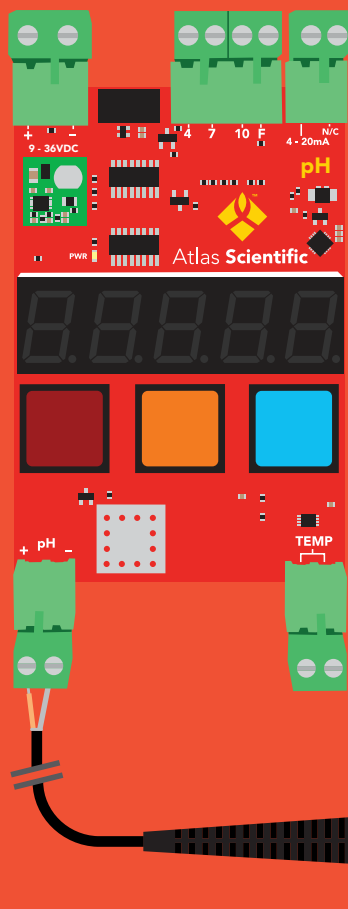
How to connect the industrial pH probe

The Atlas Scientific™ Gen 3 Industrial pH probe - No Temp, can be connected in several different ways. The following show two examples:



Using our **EZO Complete-pH™** you can easily connect the Gen 3 Industrial pH probe - No Temp, via the probes SMA connector.

Gen 3 Industrial pH probe - No Temp
with the SMA connector option.



For industrial purposes, the Gen 3 Industrial pH probe - No Temp, can be easily connected to our **Industrial pH Transmitter** via the probes tinned leads.

Gen 3 Industrial pH probe - No Temp
with the tinned leads connector option.

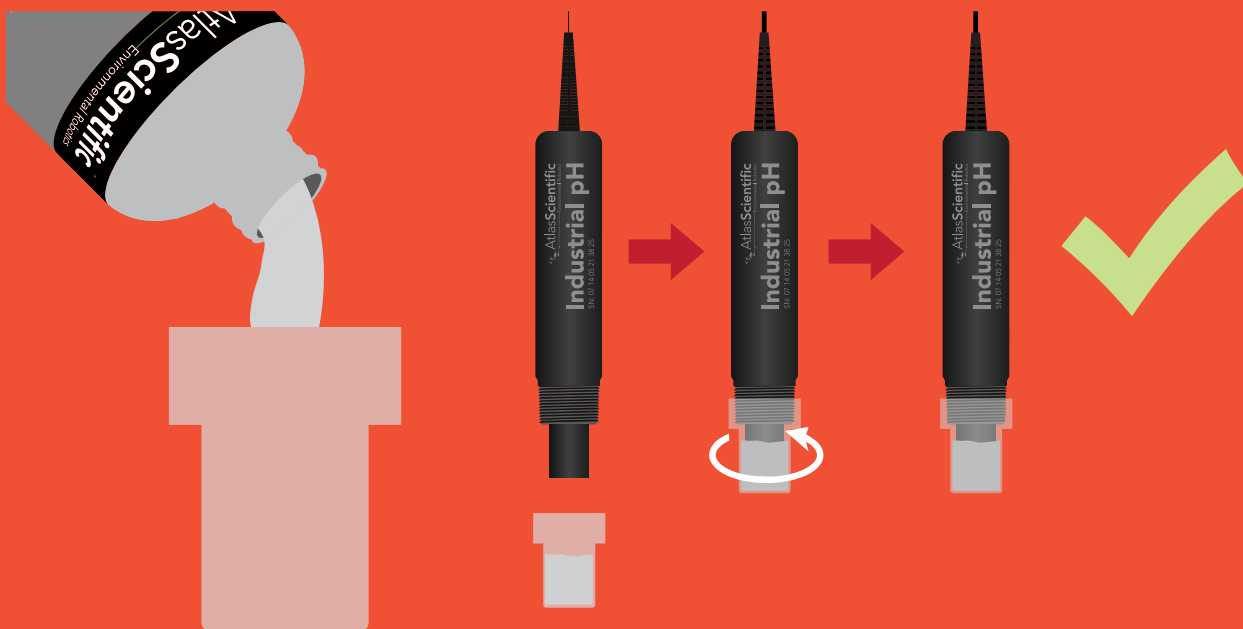
Once installed into your machine, the pH probe can be left dry for up to 3 weeks without damaging the probe, this is why every Industrial pH probe is shipped with a plastic cap containing pH probe storage solution. The cap should remain on the probe until it is used.

Remove the Industrial pH probe cap by turning it clockwise, and pulling the probe out.



Long term storage

When you are finished using the Industrial pH probe, you can prepare the probe to be used again for a later date. First, make sure the probe cap still has pH probe storage solution within it. If not, just add some from the pH probe storage solution bottle. Tighten the cap back onto the probe by turning it counterclockwise.



Probe can dry out

Unlike most pH probes, this probe can dry out. It can be left dry for up to 3 weeks without damaging the probe. However, repeated wetting and drying cycles will eventually damage the probe. For best results, keep the probe wet at all times. But if the probe is left out to dry, it should not be considered damaged. It just needs to be rehydrated.

If a pH probe is kept in a hot environment during shipping, KCl creep may cause the soaker bottle to dry out. KCl will form a salt channel that wicks the water out of the probe's soaker bottle.



Follow these instructions to rehydrate your industrial pH probe:

1. Place the probe into a beaker of tap water.
2. Let the probe sit in the water for 3–5 mins.
3. Add a scoop of KCL (*best*) or NaCl (*good*) to the beaker.
(a scoop = so much salt that no human could drink the water)
4. Let the probe sit in the salt water for 15–20 mins.
5. Done.

The probe is now ready to use.

Probe cleaning

Coating of the pH bulb can lead to erroneous readings including shortened span (slope). The type of coating will determine the cleaning technique. Soft coatings can be removed by vigorous stirring or by the use of a squirt bottle. Organic chemical, or hard coatings, should be chemically removed using a light bleach solution. If cleaning does not restore performance, reconditioning may be tried. **Do not use a brush or abrasive materials on the pH probe.**



How often do you need to recalibrate a pH probe?

Because every use case is different, there is no set schedule for recalibration.

If you are using your probe in a fish tank, a hydroponic system or any environment that has generally weak levels of acids and bases you will only need to recalibrate your probe once per year for the first two years. After that every ~six months.

If you are using the pH probe in batch chemical manufacturing, industrial process, or in a solution that is known to have strong acids and bases, then calibration should be done monthly or in extreme cases after each batch.

Probe reconditioning

When reconditioning your pH probe is required due to aging, we recommend you use the **Atlas Scientific pH probe reconditioning kit**.



1980's — Today



**Despite appearances
THE KCl CREEP
is really quite harmless.**

The white crystals
you may find on your electrode
are formed by potassium chloride (KCl)
from the electrode filling solution.
Rinse the KCl from the electrode
with distilled water and proceed as usual.



**Dried KCl residue
from pH storage
solution**

Decades later...

KCl continues to behave the same way.

If you encounter the "KCl CREEP" or, if your probe dried out during shipping; Simply rinse off your probe with water, and carry on.

Your probe is not damaged.